

A novel autonomous irrigation system for smart agriculture using AI and 6G enabled IoT network

Sitharthan R^{a,*}, Rajesh M^b, Vimal S^c, Saravana Kumar E^d, Yuvaraj S^e, Abhishek Kumar^f, Jacob Raglend I^g, Vengatesan K^b

^a School of Electrical Engineering, Vellore Institute of Technology and Science, Chennai, 600127, India

^b Department of Computer Science, Sanjivani College of Engineering, Kopergaon, India

^c Department of Computer Science Engineering, Ramco Institute of Technology, Rajapalayam, Tamilnadu, India

^d Department of Computer Science and Engineering, The Oxford College of Engineering, Bengaluru, Karnataka, India

^e Department of Electronics and Communication Engineering, SRM Institute of Science and Technology, Kattankulathur, India

^f Indian Institute of technology, Jodhpur & National institute of Design, Andhra Pradesh, India

^g Department of Electrical Engineering, School of Electrical Engineering, Vellore Institute of Technology and Science, Vellore, 632014, India

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ABSTRACT

Agriculture plays a vital role in the growth of humankind. The total world depends upon the agricultural contribution, and wherein the agriculture purely depends upon the necessity of irrigation. This paper develops an autonomous irrigation system using Artificial Intelligence (AI) and the 6G enabled Internet of Things (6G-IoT) transforming to create a hassle-free smart agriculture model. The developed autonomous irrigation system is purely based on prediction algorithm which is mount in a microprocessor, which uses weather history data using IoT to identify and predict rainfall patterns and any climatic changes; hence it creates an intelligent system which irrigates the field depending upon the outer environment. The developed system not only checks the environmental condition, but it also measures the moisture content of the soil to supply the required amount of water to the crops. The developed autonomous AI and 6G-IoT based smart agricultural irrigation system has been tested under control environment with an accuracy of 86.34%, sensitivity of 89.28% and precision of 91%. Hence, the proposed methodology could be the possible solution for autonomous irrigation of crops in future.

1. Introduction

Agriculture is probably the greatest worry to all humanity since the vast majority of food is created by farming. As of now, numerous individuals still experience the ill effects of yearning because of the absence of food in certain nations, particularly in Africa. Exceptionally, hunger caused the persistent undernourishment of in excess of 800 million individuals on the planet in 2016. All the more remarkably, in excess of 10 million people has been left to die from hunger every year [1]. Clearly, expanding food creation is without a doubt a viable method for destroying yearning and destitution. But it's not that easy, the agriculture is a long way from modernization in a large portion of agricultural nations, prompting low food creation [2].

In recent days, Smart Agriculture (SA) is an arising idea that alludes to overseeing ranches utilizing present day Information and Communication Technologies to expand the amount and nature of items while

advancing the human work required [3]. Especially, the SA frameworks bring together several advanced computer and information technologies such as Artificial Intelligence (AI), Internet of Things (IoT), and Cloud Computing (CC) into the smart agricultural creation [4]. Among these techniques, IoT and AI are the predominantly significant technique that creates the SA framework. Without the role of IoT and AI, it's significantly not possible to create a SA system [5,6]. In the SA system, the IoT is mainly used to automatically collect agricultural data and to transmit the collected data to the data centres. Whereas, artificial intelligence techniques analyze the agricultural data using the machine learning, neural network and clustering technique for smart decision-making in the SA system [7,8]. For example, when there is lack of water in the environment created with smart agriculture framework, the action will be taken immediately to watering the crops in that environment by analysing the data collected by the IoT and AI analysing the data.

Several researchers have been working to develop smart agriculture

* Corresponding author.

E-mail address: sitharthan.r@vit.ac.in (S. R).

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model using AI and IoT. In [9], smart agriculture based on IoT system and deep reinforcement learning with four-layer has been developed. The developed system also integrated AI and CC to increase smart agriculture food production. In [10], the author has developed Agri trust, (i.e.,) a trust management approach for smart agriculture using CC and IoT. The developed Agri trust identifies malicious and compromised data interval and provides a friendly agricultural environment to the people migrated from other cities. In [11], the author has developed an Agro-IoT collaboration framework for creating a smart agriculture environment. The developed Agro-IoT framework is specially designed for sugarcane agriculture. The developed framework facilitates a live communication medium between farmers and government. It also works as a monitoring medium to observe the growth of the crops, moisture content and to control the cultivation wastages. In [12], IoT and AI-based architecture system for smart cities and agriculture have been developed. The developed architecture converges the earlier system to develop a smart system using layered, service-oriented and cloud-based software architecture using IoT and the architecture enhances the yielding. In [13], the author has developed smart farming using the IoT to rectify the current food production. The author has also reviewed several methodologies addressing smart agricultural problems. The author has proved that the IoT and AI could be the methods that could solve the crop framing problems and improve the accuracy of crop diagnosis. In [14], the author investigates 6 G wireless systems with IoT for the expected support in truly global wireless communications, anywhere and anytime. The developed system with 6G-IoT is an ISTN, that is expected to be at the core of 6 G communication systems, and how to build a high-capacity yet low-cost ISTN is of significant importance in developing global 6 G communication systems. In [15], the author presents a point-by-point overview of remote advancement towards the 6G-IoT network. In this review, the great spotlight is on the new compositional changes related with 6 G networks, portrayed by universal 3D inclusion, the presentation of unavoidable AI and upgraded network convention stack. Spotted hyena optimizer, Emperor penguin optimizer with bio-inspired, Seagull optimization algorithm, STOA based bio-inspired based optimization algorithm has been developed to solve industrial engineering problems [16–20]. Later, artificial intelligence, neural network, and machine learning based solutions are been provided for complex industrial engineering problems [21–27]. These soft computing techniques can be further utilized for autonomous irrigation system. For the review, it has been identified that the autonomous irrigation system requires technology to support such as AI, 6-G network and IoT [21]. Combining all these together could build up a new paradigm that will sort out current problems in smart agriculture.

In this paper, an autonomous irrigation system for smart agriculture using Artificial Intelligence (AI) has been developed. The developed smart agriculture irrigation system helps in transforming the traditional irrigation system to create a hassle-free smart agriculture irrigation model using AI and 6 G enabled IoT. The developed autonomous irrigation system is purely based on prediction algorithm which is mount in a microprocessor, which uses weather history data using IoT to identify and predict rainfall patterns and any climatic changes; hence it creates an intelligent system which irrigates the field depending upon the outer environment. The developed system not only checks the environmental condition, but it also measures the moisture content of the soil to supply the required amount of water to the crops. The developed autonomous AI and 6 G -IoT based smart agricultural irrigation system has been tested under control environment with an accuracy of 86.34% which could probably the solution for irrigating crops in future.

2. Smart agriculture

Smart agriculture is focusses on the use of data acquired through various sources such as, geographical data, historical data and mechanical system data to manage the agricultural activities. Smart frameworks separate themselves through their capacity to record the

information and bode well out of it. Keen cultivating utilizes equipment like IoT, AI and programming Software as a service (SaaS) to catch the information and give significant bits of knowledge to deal with all the procedure on the homestead, both pre- and post-reap. The smart agriculture frame work is shown in Fig. 1. The information is coordinated, open constantly and brimming with information on each part of money and field tasks that can be observed from any place on the planet. IoT in horticulture includes sensors, robots and robots associated through the web which work consequently and semi naturally performing activities and social occasion information pointed toward expanding productivity and consistency. With expanding requests and lack of work across the globe, horticulture computerization and robots or regularly known as Agri-bots are beginning to acquire consideration among ranchers. Yield creation diminished by an expected 213 crores approx (\$3.1 billion) a year because of work deficiencies in the different countries. Ongoing headways in sensors and AI innovation that lets machines to prepare on their environmental factors have made ag-robots more prominent [28]. The world is in the beginning phases of an ag advanced mechanic's unrest with a large portion of the items still in preliminary stages and R&D mode.

In some brilliant farming system, self-loader robots with arms can identify weeds and splash pesticides in the influenced plants, setting aside the plants just as by and large pesticide costs. These robots can likewise be utilized in reaping and lifting. Hefty cultivating vehicles can likewise be explored from the solace of homes through telephone screens to perform errands and GPS can follow their situations at without fail. Robots furnished with sensors and cameras are utilized for imaging, planning and reviewing the ranches. They can be distantly controlled or they can fly consequently through programming-controlled flight plans in their inserted frameworks, working as a team with sensors and GPS. From the robot information, bits of knowledge can be drawn with respect to trim wellbeing, water system, splashing, planting, soil and field, plant checking and yield forecast and significantly more. The IoT based far off detecting uses sensors put along with the ranches like climate stations for social event information which is communicated to insightful instrument for examination.

The information gathered from light, moistness, temperature, shape and size have a possibility to change the yield. Furthermore, the information gathered by sensors regarding mugginess, temperature, dampness precipitation and dew discovery helps in deciding the climate design in ranches so development is accomplished for appropriate yields. The investigation of nature of soil helps in deciding the supplement worth and drier territories of ranches, soil seepage limit or acidity, which permits to change the measure of water required for the water system and the select most useful sort of development. PC imaging includes the utilization of sensor cameras introduced at various corners of the ranch or robots outfitted with cameras to create pictures which go through advanced picture handling. The pictures are utilized for quality control, illness identification, arranging and reviewing yield and water system observing through Image handling joined with AI which uses pictures from information base to contrast with pictures of harvests with deciding the size, shape, shading and development accordingly controlling the quality.

3. Developed AI and 6G-IoT based autonomous irrigation system

IoT based smart irrigation system framework is fit for mechanizing the water system measure by breaking down the moisture level of soil and the atmosphere condition. At the point when the power supply is given to the microcontroller, it will check the dirt moisture level content of the soil [29]. On the off chance that the moisture content isn't up to the limit then it makes the motor to pump water on consequently and turns off naturally on the off chance that it scopes to the limit level. The list of sensors applicable in the smart agriculture is listed in Table 1. The need for water for any crops is likewise decreased radically. Far off checking is too conceivable in IoT based savvy water system framework.



Fig. 1. Smart agriculture framework.

Table 1
Sensor for IoT applications in autonomous irrigation system.

Sl. No.	Sensor devices	Applications
1	Volumetric Sensor	Volumetric soil moisture sensors straightforwardly measure the measure of water in the dirt. This classification has the most sensor types: neutron dampness tests, heat dispersal sensors, and the normal, di-electric sensors.
2	Di-electric sensors	The di-electric sensors measure the di-electric steady of the dirt, an electrical property reliant on soil dampness content, and can be bought in 3 varieties: Time Domain Refractometry (TDR) sensors, Time Domain Transmissometer (TDT) sensors, and capacitance or Frequency Domain Refractometry (FDR) sensors.
3	Tensiometers	Tensiometers are soil dampness sensors that measure this strain between soil particles and water atoms. A tensiometer is a vertical, water-filled cylinder with a permeable tip that is embedded into the dirt at suggested profundities; the dirt coaxes water out of the permeable tip of the fixed cylinder, making a vacuum. Drier soils make a more grounded vacuum since water atoms are more diligently to pull off soil particles.
4	Solid state sensors	Solid state sensors are an expensive device, that is used to measure the soil data. Solid state sensors work by utilizing two cathodes to quantify the electrical obstruction in the dirt. More water in the dirt will decrease electrical opposition in light of the fact that the electrical momentum can go through the water simpler; less water will build the obstruction. After the electrical obstruction is estimated the water strain is determined to decide the plant accessible soil dampness.
5	Granular matrix sensors	Granular matrix sensors have a granular matrix inside a metal case. The electrical sensors are embedded in the granular matrix above a gypsum wafer.

3.1. IoT and AI framework

In the developed autonomous irrigation framework, there are four layers which interact with each other. The layers such as Physical, Edge, Cloud, and Network are bond with each other by means of 6G-IoT and AI through a web service. Fig. 2 shows the architecture of the developed AI and IoT based autonomous irrigation system. The web service is written in PHP with a light weighted REST API to convey the information between the field gadget and the worker. The assistance is facilitated on Apache (Web worker) at the worker machine. The microcontroller

gadget sends the field information to the worker utilizing this web administration. This web administration can deal with the organization variance/blackout during synchronizing the information from the documented gadget to the worker with the assistance of banner settings at the data set level. A web administration has been created in Arduino to gather the climate determining information. This web administration additionally totals the climate estimating information like temperature, mugginess, shadiness, UV Index and precipitation of various web determining entryways like Open Weather and Accuracy Weather utilizing API of these gateways. The created web administration read the anticipated information of the predetermined area utilizing API and store it in MySQL data set at the worker, which is considered in the forecast calculation. In the contest, the IoT Edge is an administration that expands on top of the IoT Hub and empowers clients to complete edge registering. Edge processing is when information is dissected on gadgets, that is, at the edge of the organization, as opposed to in the actual cloud. With edge processing, you can abstain from moving crude information via doing information cleaning, total, and examination on the actual gadget, and afterwards send the bits of knowledge acquired to the cloud. This will bring about diminished transmission capacity costs, faster reaction times and decreased traffic. This runs on each IoT Edge device and manages the modules deployed there. It performs a number of functions including monitoring and managing the device and facilitating communications between the modules, the device, other devices, and the cloud. The runtime supports both Linux and Windows operating systems.

The developed framework is also used to predict the soil moisture based on field information and climate determining information utilizing support AI calculation. The algorithm shows data with respect to soil moisture of the forthcoming days. It likewise gives water system recommendations, in view of the characterized level of soil moisture content and anticipated precipitation, to save water and energy. The created data by calculation and gadget is put away in MySQL Database at the worker. Besides, the created system controls the hand-off change to begin/stop the water engine. In this module, a water siphon is associated with a hand-off switch that is constrained by a Wi-Fi empowered hub. The hub is constrained by the web administration through a trigger from the responsive electronic interface for continuous observing. Utilizing this electronic interface, the water siphon can be overseen distantly in manual and auto modes. The correspondence advances utilized in the proposed design is the Wi-Fi module/6G-Mobile information correspondence. In this trial examination, a Wi-Fi module has been utilized to send information to the worker. The algorithm used in the micro

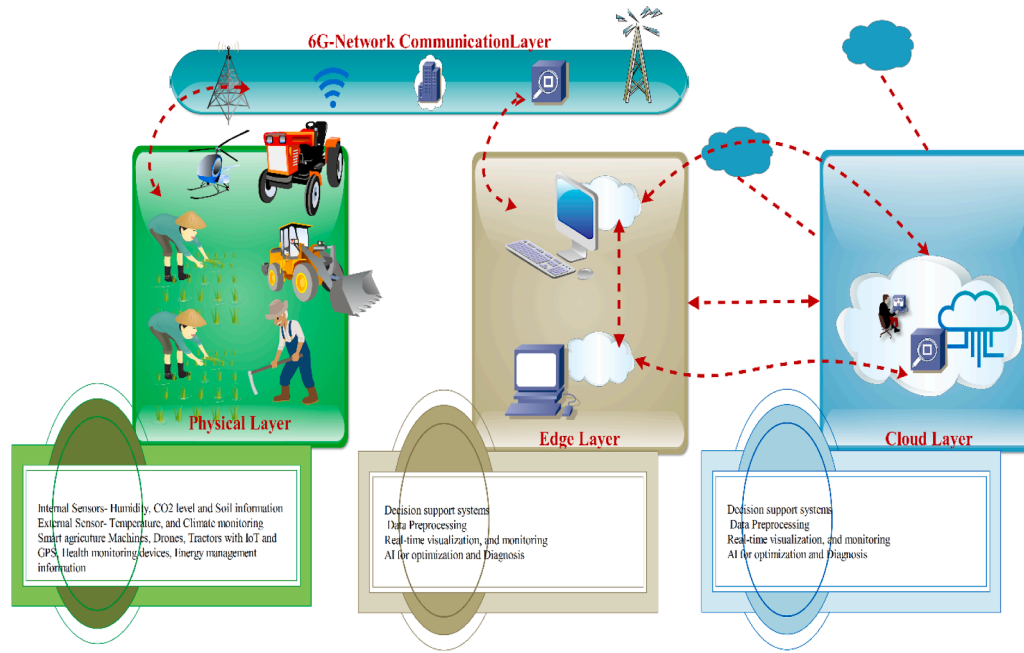


Fig. 2. AI and IoT based autonomous irrigation system framework.

controller is given below.

Algorithm of irrigation scheduling

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Initialize soil moisture content to start irrigation, ( $M_{min}$ ) or stop irrigation  $M_{max}$ 
Initialize manual/auto mode for irrigation // using 6G based IoT or manual method
for irrigation
If mode=auto
{
  Process soil moisture content ( $S_m$ )
  If ( $S_m \leq M_{min}$ )
  {
     $N_n$  = Climate and Weather data obtained from network layer
    Optimize required soil moisture ( $SO_m$ ) using AI techniques based on the  $S_m$ ,  $M_{min}$ 
    and  $N_n$  data. // Measure the soil moisture condition
    Set  $M_{max}$  = least ( $M_{min} + SO_m + N_n$ ) // amount of soil required for the field
    while ( $M_{max} > S_m$ ) // check the moisture content of the soil.
    {
      Motor=1 to relay switch // Motor pump gets ON to irrigate the field
    }
    Motor=0 to relay switch // Motor pump gets OFF to stop irrigation of the field
  }
  Else
  {
    Motor=0 to relay switch // Moto pump gets OFF when maximum moisture content
    reached
  }
}
Else
{
  Scheduled irrigation // mode=auto
  Set irrigation date= $I_d$ 
  If (calendar date  $\geq I_d$ ) // condition to check the scheduled date
  {
    while ( $M_{max} > S_m$ ) // check the moisture content of the soil
    {
      Motor=1 to relay switch // Motor pump gets ON to irrigate the field
    }
  }
  Else
  {
    Motor=0 to relay switch // Motor pump gets OFF to stop irrigation of the field
  }
}
}

```

3.1.1. AI-Enabled intelligent 6G networks

AI techniques provide the best opportunity to train auto-learning models to realize network optimization for 6 G wireless networks, allowing providers or operators to optimize the network parameters, resources or architectures to better adapt services for devices, finally make 6 G networks become intelligent, and agile. The improvement of 6 G network has an enormous scope, multi-layered, high perplexing, dynamic, and heterogeneous. Moreover, 6 G network need to help consistent availability and assurance different QoS necessities of the tremendous number of gadgets, just as cycle enormous measure of information produced from actual conditions. Computer-based intelligence procedures with ground breaking examination capacity, learning capacity, upgrading capacity and astute acknowledgement capacity, which can be utilized into 6 G network to brilliantly do execution advancement, information revelation, modern learning, structure association and confounded dynamic [30,31]. With the assistance of AI, we present an AI-empowered smart design for 6 G network which is fundamentally isolated into four layers as illustrated in Fig. 2.

3.2. Experimental setup

The main objective of this experiment is to collect the physical parameters of a farming land using sensors and to utilize these sensors' data along with weather forecast information for developing an algorithm for prediction of soil moisture of the upcoming days. The algorithm uses a hybrid machine learning approach, as discussed in Section 3.1, to achieve higher accuracy in soil moisture prediction. The experimental setup is shown in Fig. 3. The Fig. 3(a) illustrates the architecture of the control connection kit, which consists of 6G-communication module with IoT enabled for fetching weather data and climate change data. The microcontroller board is connected to the tensiometer sensor. The tensiometer sensor is a filed data collection device that has been deployed in the garden of the test environment which was created to assess the performance of the autonomous irrigation system using AI-IoT based 6G network. The data collected in the test environment is stores in the server using the web service. Later, the collected data are been analysed using the AI approach and 6G-IoT is responsive for the real time monitoring and data visualization. Fig. 3(c) depicts the test

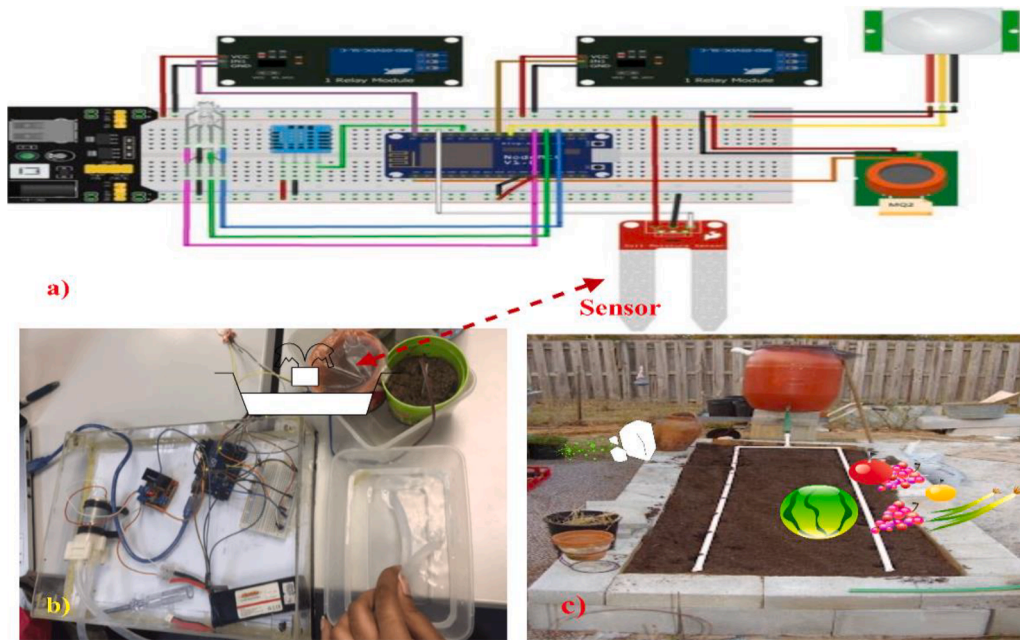


Fig. 3. Experimental setup.

environment developed for assessing the proposed autonomous irrigation system using AI-IoT based 6G network.

Fig. 3(b) shows the 6G-IoT of water system arranging module to plan the water system by setting Irrigation Date and Threshold Levels for Soil Moisture. This data, anticipated soil dampness and precipitation data will help client/rancher in arranging/planning of ideal water system. The created module works in two modes, viz., auto, and manual. In manual mode, the client takes the planning choice dependent on anticipated soil dampness and precipitation data. In auto mode, the client sets the Soil Moisture Threshold levels, and the framework naturally plans the water system date dependent on the anticipated soil dampness and climate conjecture (precipitation) data. For instance, if the downpour is normal on the planned water system date or close to the planned date, at that point, the framework will hang tight for the downpour and it will suspend the counterfeit/manual water system or it will begin watering (if necessary, in light of the calculation) the field to keep up least soil dampness work the appearance of the downpour. Further, the framework can deal with the progressions in gage precipitation esteems. The developed prototype can be extended to irrigate

large area.

4. Result and discussion

A profound assessment has been carried out to evaluate the performance of proposed autonomous irrigation system using AI-IoT based 6G network. In this assessment the developed module as discussed in Section 3.2 is assessed in the test environment created with devices like halogen lamp, measured oxygen content, air conditioner and water filter. The devices could absorb the moisture content of the soil and will help for fast testing. The testing was conducted on 24/12/2020 and 25/12/2020. The temperature of the test environment was measured in the interval of an hour.

The halogen lamp in the test environment has been used like the daylight, the intensity of luminous has been increased during day bright hours. The data observed for the room temperature is depicted in the Fig. 4. From the figure it can be observed that the test environment temperature varies from 19°C to 33 °C. Further, in the meantime, humidity level has been assessed in the test environment and obtained

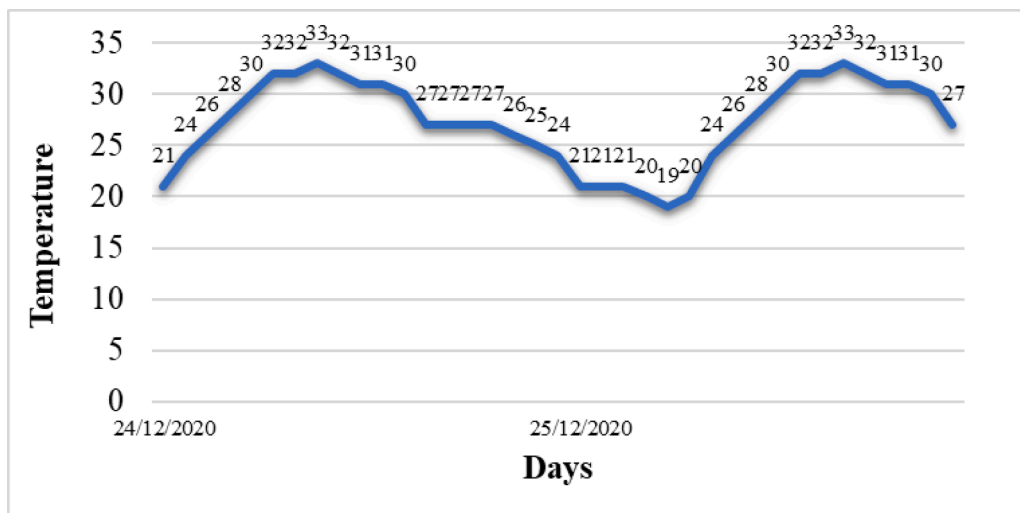


Fig. 4. Temperature observed in the test environment.

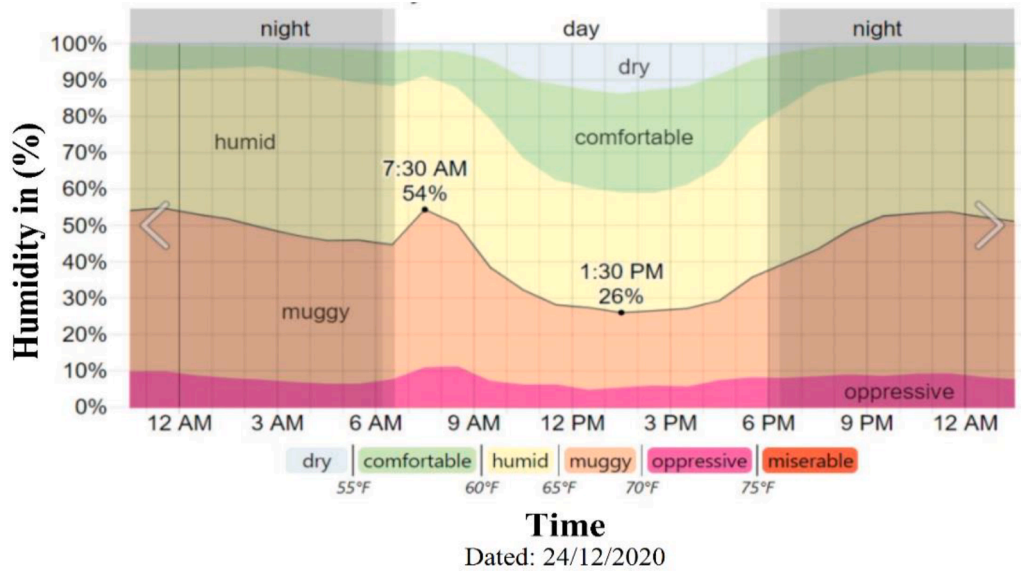


Fig. 5. Humidity observed in the test environment.

humidity measurement is depicted in Fig. 5. From the figure it can be observed that humidity is 55% during morning and evening, and is 24% to 26% during day time. Moreover, the moisture level of the soil has been measured using, Volumetric Sensor, Tensiometer sensor, and Solid-state sensor and fed with the 6G-IoT module and AI system for further processing. The moisture level varies between 10% to 45% and the observation data obtained for the test environment is depicted in the Fig. 6. From the figure it has been observed that the soil moisture reduces with high light luminous and increases with low luminous.

The developed autonomous irrigation system using AI-IoT based 6G network is engaged with the auto mode. The module has collected the soil moisture content and climate data using the 6G enabled IoT server network. Meanwhile, it also feeds the soil moisture and temperature data obtained using the sensor into the server using the 6G enabled IoT module. Further, the data is processed by the AI and motor pump is turned ON/OFF according to the requirement. Fig. 7 depicts the water irrigation data log obtained in percentage of engaging based on the soil moisture content. From the figure it can be observed that the motor

pump operates for 0 to 75% based on the variation of soil moisture (soil moisture is varied using halogen lamp operation in the test environment). The AI process in motor operation can be clearly identified in the Fig. 7, when the moisture level lowers 20% the motor pump is engaged to water. Using the developed autonomous irrigation system using AI-IoT based 6G network the farmers could monitor the water flow level and can also prevent over watering. Further, in the investigation the performance evaluation of different sensor used in the study has been evaluated. Fig. 8 shows the performance evaluation of the different sensor engaged in the analysis. In this analysis, Tensiometer sensor, and Solid-state sensor has higher performance with a accurate date precision.

Further, in the assessment, the water irrigation prediction and controlling are further analysed based on the accuracy, sensitivity, and precision. Where, accuracy is the percentage of the example set that is effectively characterized in the exactness of a classifier, as given in underneath condition,

$$Accuracy = \frac{Number\ of\ accurate\ positive + Number\ of\ accurate\ negative}{Number\ of\ accurate\ positive + faulty\ positive + faulty\ negative + accurate\ negative} \quad (1)$$

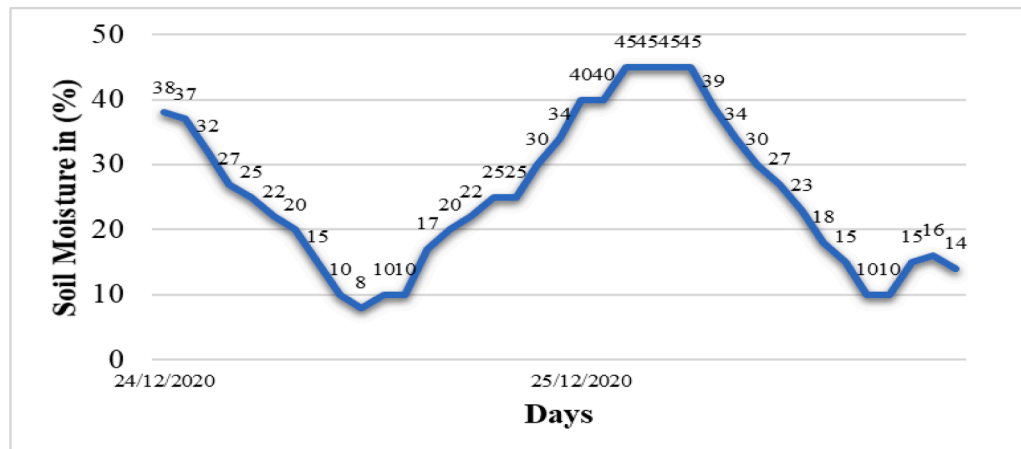


Fig. 6. Soil moisture observed in the test environment.

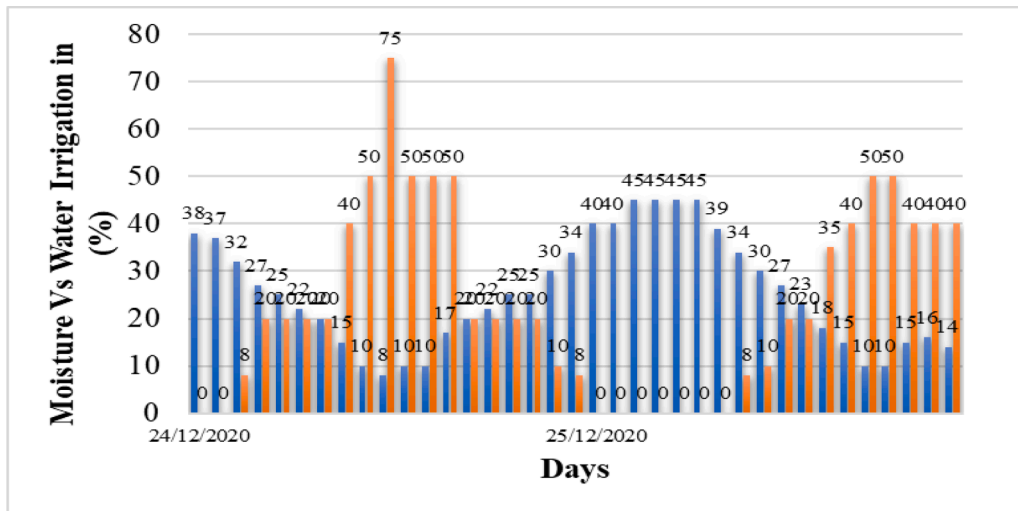


Fig. 7. Function of proposed autonomous irrigation system using AI-IoT based 6G network.

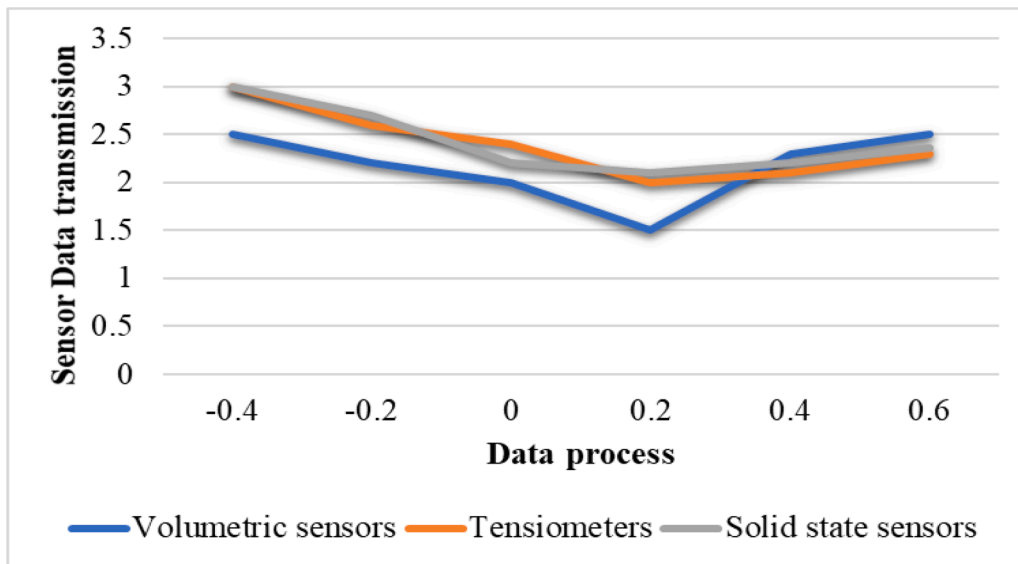


Fig. 8. Performance evaluation between sensors.

Sensitivity is the extent of positive tuples that are effectively recognized, which is otherwise called Genuine Positive Rate given by condition,

$$\text{Sensitivity} = \frac{\text{Number of accurate positive}}{\text{Number of accurate positive} + \text{Number of faulty negative}} \quad (2)$$

Precision is the extent of genuine positives against all the positive outcomes and the scientific portrayal of which is given

$$\text{Precision} = \frac{\text{Number of accurate positive}}{\text{Number of accurate positive} + \text{Number of faulty positive}} \quad (3)$$

Table 2

Accuracy, sensitivity and precision of the various methods for autonomous irrigation system using AI-IoT based 6G network.

Methods	Accuracy	Sensitivity	Precision
Proposed	86.34%	89.28%	91%
LR with IoT	83.21%	84.32%	89%
AI with web service	81%	89%	88%
ANFIS	72%	83.39%	87%

The assessment data of accuracy, sensitivity and precision of the various methods for autonomous irrigation system using AI-IoT based 6G network is listed in Table 2. From the observation it can be determined that proposed algorithm has an accuracy of 86.34% when compared to other methods. Further, it has a sensitivity of 89.28% with higher sensitivity than the other competitive method. It also has a precision of 91% which is higher precision value than the competitive method. Fig. 9 depicts the Performance analysis of different technique developed for autonomous irrigation. From the analysis the developed system not only checks the environmental condition, but also measures the moisture content of the soil to supply the required amount of water to the crops. It also makes a faster communication between farmers and cloud data during irrigation. Therefore, the developed framework is an effective methodology in irrigating the crops.

5. Conclusion

In this work, an autonomous irrigation system for smart agriculture using Artificial Intelligence (AI) and 6G enabled IoT has been developed. The developed 6G-IoT and AI based system helps in transforming the

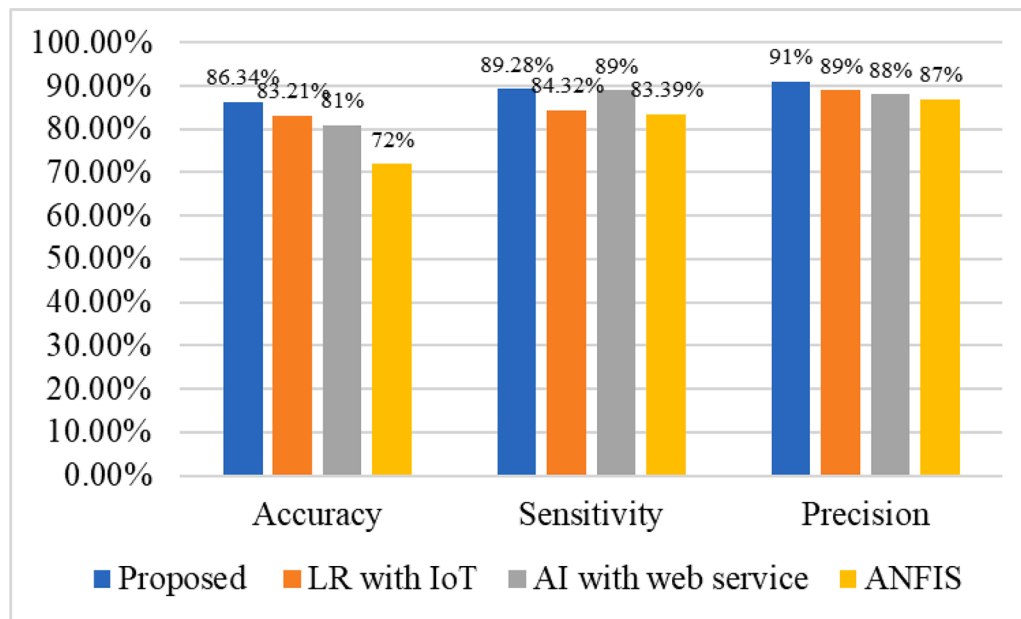


Fig. 9. Performance analysis of different technique developed for autonomous irrigation.

traditional irrigation system to create a hassle-free smart agriculture irrigation model. The developed autonomous AI and 6G-IoT based smart agricultural irrigation system has been tested under control environment and following observation has been identified,

- The developed autonomous irrigation system has an accuracy of 86.34%.
- The developed system not only checks the environmental condition, but also measures the moisture content of the soil to supply the required amount of water to the crops.
- The developed system communicates with the farmers regarding the weather history and helps in predicting the rainfall pattern through analysing the weather report and climate changes.

The developed framework is an effective methodology in irrigating the crops. The developed framework works in all environmental conditions and could be the first step in developing smart agriculture system.

Declaration of Competing Interest

The authors declare that they didn't get any financial support or influential support to be reported in this paper.

References

- [1] J. Clover, Food security in sub-Saharan Africa, *Afr. Security Stud.* 12 (1) (2003) 5–15.
- [2] T. Talhelm, A.S. English, Historically rice-farming societies have tighter social norms in China and worldwide, *Proceed. Nat. Acad. Sci.* 117 (33) (2020) 19816–19824.
- [3] A. Escamilla-García, G.M. Soto-Zarazúa, M. Toledano-Ayala, E. Rivas-Araiza, A. Gastélum-Barrios, Applications of artificial neural networks in greenhouse technology and overview for smart agriculture development, *Appl. Sci.* 10 (11) (2020) 3835.
- [4] D. Garg, M. Alam, Deep Learning and IoT For Agricultural Applications. In *Internet of Things (IoT)*, Springer, Cham, 2020, pp. 273–284.
- [5] A.D. Boursianis, M.S. Papadopolou, P. Diamantoulakis, A. Liopa-Tsakalidi, P. Barouchas, G. Salahas, G. Karagiannidis, S. Wan, S.K. Goudos, Internet of things (IoT) and agricultural unmanned aerial vehicles (UAVs) in smart farming: a comprehensive review, *Internet of Things* (2020), 100187.
- [6] O. Debauche, S. Mahmoudi, M. Elmoulat, S.A. Mahmoudi, P. Manneback, F. Lebeau, Edge AI-IoT pivot irrigation, plant diseases, and pests identification, *Procedia Comput. Sci.* 177 (2020) 40–48.
- [7] L. García, L. Parra, J.M. Jimenez, J. Lloret, P. Lorenz, IoT-based smart irrigation systems: an overview on the recent trends on sensors and IoT systems for irrigation in precision agriculture, *Sensors*, 20 (4) (2020) 1042.
- [8] P. Paul, M.R. Saavedra, P.S. Aithal, R.R.S. Ripu Ranjan Sinha, P.S.B. Aremu, Agro informatics Vis-à-Vis internet of things (IoT) integration & potentialities—an analysis, *Agro Economist-An Int. J.* 7 (1) (2020) 13–20.
- [9] F. Bu, X. Wang, A smart agriculture IoT system based on deep reinforcement learning, *Future Generation Comput. Syst.* 99 (2019) 500–507.
- [10] K.A. Awan, I. Ud Din, A. Almogren, H. Almajed, AgriTrust—A trust management approach for smart agriculture in cloud-based internet of agriculture things, *Sensors* 20 (21) (2020) 6174.
- [11] M. Pachayappan, C. Ganeshkumar, N. Sugundan, Technological implication and its impact in agricultural sector: an IoT based collaboration framework, *Procedia Comput. Sci.* 171 (2020) 1166–1173.
- [12] N. Gavrilović, A. Mishra, Software architecture of the internet of things (IoT) for smart city, healthcare and agriculture: analysis and improvement directions, *J. Ambient Intell. Humaniz Comput.* (2020) 1–22.
- [13] E. Navarro, N. Costa, A. Pereira, A systematic review of IoT solutions for smart farming, *Sensors* 20 (15) (2020) 4231.
- [14] X. Huang, J.A. Zhang, R.P. Liu, Y.J. Guo, L. Hanzo, Airplane-aided integrated networking for 6G wireless: will it work? *IEEE Vehicular Technol. Magazine* 14 (3) (2019) 84–91.
- [15] W. Saad, M. Bennis, M. Chen, A vision of 6G wireless systems: applications, trends, technologies, and open research problems, *IEEE Netw.* 34 (3) (2019) 134–142.
- [16] G. Dhiman, V. Kumar, Spotted hyena optimizer: a novel bio-inspired based metaheuristic technique for engineering applications, *Adv. Eng. Softw.* 114 (2017) 48–70.
- [17] G. Dhiman, V. Kumar, Emperor penguin optimizer: a bio-inspired algorithm for engineering problems, *Knowl. Based Syst.* 159 (2018) 20–50.
- [18] G. Dhiman, V. Kumar, Seagull optimization algorithm: theory and its applications for large-scale industrial engineering problems, *Knowl. Based Syst.* 165 (2019) 169–196.
- [19] S. Wolfert, L. Ge, C. Verdouw, M.J. Bogaardt, Big data in smart farming—a review, *Agric. Syst.* 153 (2017) 69–80.
- [20] G. Dhiman, A. Kaur, STOA: a bio-inspired based optimization algorithm for industrial engineering problems, *Eng. Appl. Artif. Intell.* 82 (2019) 148–174.
- [21] A. Walter, R. Finger, R. Huber, N. Buchmann, Opinion: smart farming is key to developing sustainable agriculture, *Proceed. Nat. Acad. Sci.* 114 (24) (2017) 6148–6150.
- [22] X. Lin, X. Sun, G. Manogaran, B.S. Rawal, Advanced energy consumption system for smart farm based on reactive energy utilization technologies, *Environ. Impact Assess. Rev.* 86 (2021), 106496.
- [23] N. Deepa, K. Srinivasan, C.Y. Chang, A.K. Bashir, An efficient ensemble vtopes multi-criteria decision-making model for sustainable sugarcane farms, *Sustainability* 11 (16) (2019) 4288.
- [24] R.G. Babu, M.S. Obaidat, V. Amudha, R. Manoharan, R. Sitharthan, Comparative analysis of distributive linear and non-linear optimised spectrum sensing clustering techniques in cognitive radio network systems, *IET Networks* (2020).
- [25] R. Sitharthan, S. Krishnamoorthy, P. Sanjeevikummar, J.B. Holm-Nielsen, R.R. Singh, M. Rajesh, Torque ripple minimization of PMSM using an adaptive Elman neural network-controlled feedback linearization-based direct torque control strategy, *Int. Trans. Electric. Energy Syst.* (2020), <https://doi.org/10.1002/2050-7038.12685>.

- [26] S.K. Nataraj, F. Al-Turjman, A.H. Adom, R. Sitharthan, M. Rajesh, R. Kumar, Intelligent robotic chair with thought control and communication aid using higher order spectra band features, *IEEE Sens. J.* (2020), <https://doi.org/10.1109/JSEN.2020.3020971>.
- [27] S. Kaur, L.K. Awasthi, A.L. Sangal, G. Dhiman, Tunicate swarm algorithm: a new bio-inspired based metaheuristic paradigm for global optimization, *Eng. Appl. Artif. Intell.* 90 (2020), 103541.
- [28] G. Dhiman, M. Garg, A. Nagar, V. Kumar, M. Dehghani, A novel algorithm for global optimization: rat swarm optimizer, *J. Ambient Intell. Humaniz Comput.* (2020) 1–26.
- [29] M. Dehghani, Z. Montazeri, G. Dhiman, O.P. Malik, R. Morales-Mendez, R. A. Ramirez-Mendoza, A. Dehghani, J.M. Guerrero, L. Parra-Arroyo, A spring search algorithm applied to engineering optimization problems, *Appl. Sci.* 10 (18) (2020) 6173.
- [30] G. Dhiman, ESA: a hybrid bio-inspired metaheuristic optimization approach for engineering problems, *Eng. Comput.* (2019) 1–31.
- [31] D. Elavarasan, D.R. Vincent, V. Sharma, A.Y. Zomaya, K. Srinivasan, Forecasting yield by integrating agrarian factors and machine learning models: a survey, *Comput. Electron. Agriculture* 155 (2018) 257–282.



Dr. R. Sitharthan received his B.E. degree in Electrical and Electronics Engineering, M.E. degree in Power Systems Engineering, and Ph.D. degree in Electrical Engineering from the Anna University, India, in 2010, 2012, and 2016, respectively. He is an assistant professor in the School of Electrical Engineering, Vellore Institute of Technology, Vellore, Tamil Nadu, India. He has completed a research funded project as a principal investigator under the ECRA scheme, Science and Engineering Research Board, Department of Science and Technology, Government of India. His research interests include renewable energy systems, artificial intelligence-based control methodology, FACTS devices, IoT applications soft computing techniques, and piezoelectric materials.



Dr. M. Rajesh is working as associate professor in the computer science department of Sanjivani College of Engineering, Kopergaon, India, and he is also a research associate in Raga Academic Solutions, India. He is Associate Editor in 3D Research (Springer) and Editor in Mathematical and Computational Forestry & Natural-Resource Sciences (MCFNS). He has published over 60 papers in International refereed journals like IEEE, Springer and Elsevier. He Served as a reviewer for Springer, Inderscience and Elsevier journals. He is a regular reviewer of Wireless Personal Communications (Springer). In addition to it, he has reviewed many publications for SCI and Scopus indexed journal.



Dr. S. Vimal is working in Department of Information Technology, National Engineering College, Kovilpatti, Tamilnadu, India. He has around Thirteen years of teaching experience, EMC certified Data science Associate and CCNA certified professional too. He holds a Ph.D in Information and Communication Engineering from Anna University Chennai and he received Masters Degree from Anna University Coimbatore. He is a member of various professional bodies and organized various funded workshops and seminars. He has wide publications in the highly impact journals in the area of Data Analytics, Networking and Security issues and published 04 book chapters. He has hosted two special session for IEEE sponsored conference in Osaka, Japan and Thailand. He has acted as

Session chairs, organizing committee member, advisory committee and outreach committee member in various international conferences in IEEE and Other prestigious Conferences. His areas of interest include Game Modelling, Artificial Intelligence, Cognitive radio networks, Network security, Machine Learning and Big data Analytics. He is a Senior member in IEEE and holds membership in various professional bodies. He has served as reviewer for Springer, Elsevier and Wiley journals. He has hosted 9 special issues in Elsevier, Springer and CMC tech science journals. He has served as Guest editor for SCI journals and edited 3 books in scopus indexed. He is a Senior Member in IEEE.



Dr. E. Saravana Kumar is currently working as Professor in the Department of Computer Science and Engineering, The Oxford College of Engineering Bengaluru, Karnataka. He received Doctor of Philosophy Degree in Information and Communication Engineering from Anna University-Chennai, Master of Engineering Degree in Computer Science and Engineering from Annamalai University - Chidambaram and Bachelor of Engineering Degree in Electronics and Communication Engineering from Bharatiar University-Coimbatore. He has 17 years of teaching experience. He is the author of two text books. He has two patents and a good number of publications in peer-reviewed reputed journals and conferences. He has received funded projects from AICTE and VGST. His research interest includes Grid computing, Cloud Computing, IoT, Image processing and Data Mining. He is a member of IEEE, ISTE, Institution of Engineers India and International Association of Engineers.



Dr. S. Yuvaraj currently working as Assistant Professor in the Department of Electronics and Communication Engineering, SRM Institute of Science and Technology, Kattankulathur. He has completed his BE-ECE from Anna University in the year 2008 and his M. Tech-VLSI Design from SRM Institute of Science and technology in the year 2010 and received his Ph. D in the field of vlsi cyber physical security on 2019. Having more than 20 papers in indexed journals.



Dr. Abhishek Kumar is working as an Assistant Professor at School of Computer Science & IT, JAIN (Deemed to be University) and former Assistant Professor, Department of Computer Science, Institute of Science at Banaras Hindu University. He has awarded Ph.D., Doctorate in Computer Application (Research Interests: Stereoscapy, 3D Animation, Design, Computer Graphics & visual effects) and Dual Master Degree in Computer Science and Animation. Dr. Abhishek Kumar is Apple Certified Associate (USA), Adobe Certified Educator (USA) & Certified by Autodesk. Dr. Abhishek holding 8 National Patent & One International Granted in the field of VR and IOT. Dr. Abhishek has Published 40+ Research papers Scopus/WOS indexed & authored two books "Beginning PBR Texturing" & "VR Integrated Heritage Recreation" Published at Apress Springer Nature. He has trained over 75,000+ students across the globe from 153 Countries, top 5 countries are India, Germany, United States, Spain & Australia.



Dr. I. Jacob Raglend received his Bachelors degree in Electrical Engineering from The Indian Engineering College and the Masters degree in Power Systems Engineering from Annamalai University with first class in 2000 and 2001 respectively. He has done his Ph.D. degree in the Department of Electrical and Electronics Engineering, Indian Institute of Technology, Roorkee, India in the year 2007. Presently he is working as a Professor and Head Electrical Engineering Department in the School of Electrical Engineering, Vellore Institute of Technology (VIT) India. He has published around 80 papers in referred Journals and contributed around 65 papers in various national and international conferences and published 13 chapters in edited books. 12 PhD scholars have completed their Doctoral degree from him and 6 are in progress. Being a Technical and National Advisory member in various National and International Conferences. He was the publication chair for 5 IEEE International conferences and Organizing Chair for 2 IEEE International Conference. Actively involved in the students Nano Satellite project as Associate Project Director at Noorul Islam University. Being the reviewer of research articles in the Journal of Applied Soft Computing, International Journal of Energy Sector Management, International Journal of Electric Power and Energy Systems. He taught course in Basic Electrical Engineering, Power Systems, Artificial Intelligence and Soft Computing Techniques. His field of interest is Unit Commitment, Economic Dispatch, Smart Grid, Power System Restructuring and Deregulation, Renewable Energy Integration, IoT Applications, Artificial Intelligence Applications to Power System and FACTS.



Dr. K. Vengatesan currently working as Professor at the Department of Computer Engineering, Sanjivani College of Engineering, 15 years of teaching experience in computer science engineering. He received a Ph.D. Computer Science and Engineering, SSSUTMS, Bhopal 2017, pursued M.Tech in Information Technology (2008–2010) from Sathyabama University, Chennai, B.E in Computer Science Engineering (2001–2005) in PGP College of Engineering And Technology (Anna University Chennai) Namakkal. His-research area is in Data mining, bioinformatics, clustering, correlation& Bi-clustering. Life Time Member of Indian Society for Technical Education. Reviewer following journals International Journal of Medical Engineering and Informatics (IJMEI), Interscience, CONCURRENCY AND COMPUTATION: PRACTICE AND EXPERIENCE, Progress of Electrical and Electronic Engineering, which Publishing Pte. Ltd.